

UNIT-1

1. Explain about asymptotic notations. (you have to explain about time complexity, space complexity, big oh, big omega, theta notations)
2. Linear search and Binary search (you can expect one out of these two you need to write algorithm, time complexity and example. More preference for binary search)

(e three questions lo okati compulsory in sem and in first mid)

3. Stack definition. Stack ADT. Operations of stack (push operation, pop operation) functions
4. Infix to postfix conversion algorithm and example
5. Postfix expression evaluation algorithm and example

(e three questions lo okati compulsory in sem and in first mid)

Unit-2

1. queue definition, queue ADT. queue operations (insertion, delete, display functions)
2. circular queue operations (insertion , deletion, display functions)
3. Dequeue operations (front insertion, front deletion, rare insertion, rare deletion, display)
4. difference between linear queue and circular queue / linear queue and dequeue

(2 questions you can get from these three in sem and in mid)

UNIT-3

1. Single linked list operations (creation, insertion, deletion, search, display operations)
2. Stack or queue implementation using single linked list
3. Circular single linked list operations (all operations)
4. Double linked list operations (all operations)

(more expected question is single linked list)

Unit-4

1. Creation of binary search tree, insertion and deletion operation in binary search tree.
2. AVL tree creation, insertion and deletion (more important AVL tree rotations, insertion and deletion)
3. B-tree creation, insertion and deletion operations.
(one question in sem and 2 questions in mid you can expect)

Unit-5

1. Short notes on priority queues.
2. Sorting mergesort, quick sort, radix sort and heap sort
(more important quick sort. You can expect quick or merge in sem exam)
(one question in sem and 2 questions in mid you can expect)

UNIT-6

1. DFS or BFS algorithm and example (more expected question)
2. Hashing methods (division method, Collision Resolution Techniques-Separate Chaining, open addressing-linear probing, quadratic probing, double hashing, Rehashing)
Any technique you can expect